

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2022 P1H Worked Solutions

Jan 2022 | P1H

29 questions ■ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

PAST PAPER QUESTION**Q#: 1****Marks: 6****TOPIC: Algebraic Roots & Indices**

Jan 2022 P1H - Jan 2022 | P1H

1 (a) Simplify $a^7 \times a^4$
(1)(b) Simplify $w^{15} \div w^3$
(1)(c) Simplify $(8x^5y^3)^2$
(2)(d) Make t the subject of $c = t^3 - 8v$
(2)

(Total for Question 1 is 6 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 6

TOPIC: **Algebraic Roots & Indices**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Ratio Problem Solving**

Jan 2022 P1H - Jan 2022 | P1H

- 2** Danil, Gabriel and Hadley share some money in the ratios 3:5:9
The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros.

Work out the amount of money that Danil receives.

..... euros

(Total for Question 2 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 2

Marks: 3

TOPIC: **Ratio Problem Solving**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

- 3** The diagram shows triangle ABC

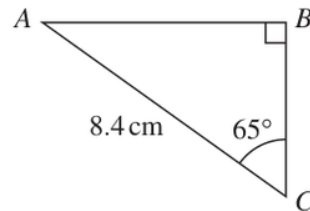


Diagram **NOT**
accurately drawn

Work out the length of the side AB
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3

Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 5****TOPIC: Percentages**

Jan 2022 P1H - Jan 2022 | P1H

4 Sarah makes and sells mugs.

One day she makes 150 mugs.

Her total cost for making these mugs is £1140

Of these mugs

 $\frac{2}{5}$ are small mugs

32% are medium mugs

and the rest are large mugs

Here is Sarah's price list for selling each mug.

MUGS	
Small	£8.50
Medium	£11.20
Large	£14.20

Sarah sells all 150 mugs.

Work out her percentage profit.

Give your answer correct to the nearest whole number.

.....%

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 5

TOPIC: **Percentages**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 5****Marks: 6****TOPIC: Statistics Toolkit**

Jan 2022 P1H - Jan 2022 | P1H

5 Jenny has six cards.

Each card has a whole number written on it so that

the smallest number is 5

the largest number is 24

the median of the six numbers is 14

the mode of the six numbers is 8

Jenny arranges her cards so that the numbers are in order of size.

5					24
---	-------	-------	-------	-------	----

(a) For the remaining four cards, write on each dotted line a number that could be on the card.

(3)

A basketball team plays 6 games.

After playing 5 games, the team has a mean score of 21 points per game.

After playing 6 games, the team has a mean score of 23 points per game.

(b) Work out the number of points the team scored in its 6th game.

(3)

(Total for Question 5 is 6 marks)

PAST PAPER SOLUTION

Q#: 5

Marks: 6

TOPIC: **Statistics Toolkit**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 5****TOPIC: Factorising**

Jan 2022 P1H - Jan 2022 | P1H

6 (a) Solve the inequality $5x - 7 \leq 2$
(2)(b) (i) Factorise $y^2 - 2y - 35$
(2)(ii) Hence, solve $y^2 - 2y - 35 = 0$
(1)

(Total for Question 6 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 6

Marks: 5

TOPIC: **Factorising**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

Jan 2022 P1H - Jan 2022 | P1H

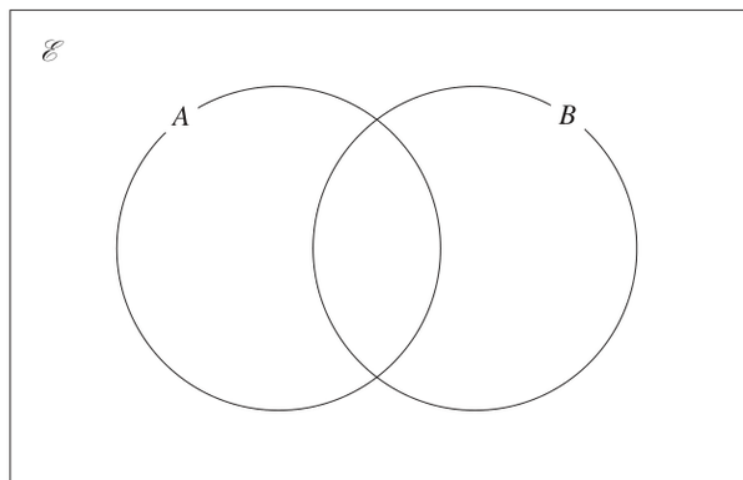
7 $\mathcal{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

$$A \cap B = \{5, 10, 15\}$$

$$B' = \{7, 8, 9, 11, 12, 13, 14\}$$

$$A' = \{4, 6, 7, 8, 14\}$$

Complete the Venn diagram for this information.



(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7

Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 2****TOPIC: Powers, Roots & Standard Form**

Jan 2022 P1H - Jan 2022 | P1H

8

$$a = 4.2 \times 10^{-24}$$

$$b = 3 \times 10^{145}$$

Work out the value of $a \times b$

Give your answer in standard form.

(Total for Question 8 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 8

Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 9****Marks: 4****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jan 2022 P1H - Jan 2022 | P1H

- 9 The diagram shows isosceles triangle ABC

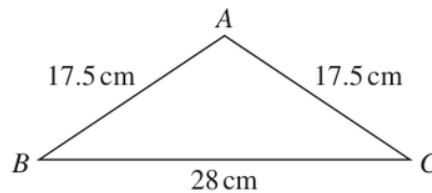


Diagram **NOT**
accurately drawn

$$AB = AC = 17.5 \text{ cm}$$

$$BC = 28 \text{ cm}$$

Calculate the area of triangle ABC

..... cm^2

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 4

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Coordinate Geometry**

Jan 2022 P1H - Jan 2022 | P1H

10 The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

.....
(2)

(b) Find the coordinates of the point where **L** crosses the y-axis.

(.....,)
(1)

(Total for Question 10 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 10

Marks: 3

TOPIC: **Coordinate Geometry**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 11****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2022 P1H - Jan 2022 | P1H

11 Himari invests 200000 yen for 3 years in a savings account paying compound interest.

The rate of interest is 1.8% for the first year and $x\%$ for each of the second year and the third year.

The value of the investment at the end of the third year is 209754 yen.

Work out the value of x

Give your answer correct to one decimal place.

$x = \dots\dots\dots$

(Total for Question 11 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 11

Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 12** **Marks: 6****TOPIC: Cumulative Frequency Diagrams**

Jan 2022 P1H - Jan 2022 | P1H

12 The table gives information about the times, in minutes, taken by 80 customers to do their shopping in a supermarket.

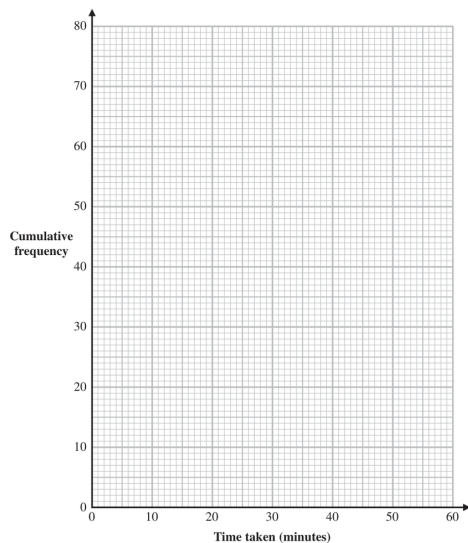
Time taken (t minutes)	Frequency
$0 < t \leq 10$	7
$10 < t \leq 20$	26
$20 < t \leq 30$	24
$30 < t \leq 40$	14
$40 < t \leq 50$	7
$50 < t \leq 60$	2

(a) Complete the cumulative frequency table.

Time taken (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median time taken.

_____ minutes
(1)

One of the 80 customers is chosen at random.

(d) Use your graph to find an estimate for the probability that the time taken by this customer was more than 42 minutes.

_____ (2)

(Total for Question 12 is 6 marks)

PAST PAPER SOLUTION

Q#: 12

Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 13****Marks: 6****TOPIC: Expanding Brackets**

Jan 2022 P1H - Jan 2022 | P1H

13 (a) Expand and simplify $5x(x + 2)(3x - 4)$
(3)(b) Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{\frac{3}{4}}$
(3)

(Total for Question 13 is 6 marks)

EA

PAST PAPER SOLUTION

Q#: 13

Marks: 6

TOPIC: **Expanding Brackets**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER SOLUTION

Q#: 14

Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Direct & Inverse Proportion**

Jan 2022 P1H - Jan 2022 | P1H

15 A is inversely proportional to C^2 $A = 40$ when $C = 1.5$ Calculate the value of C when $A = 1000$ $C = \dots\dots\dots$

(Total for Question 15 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 15

Marks: 3

TOPIC: **Direct & Inverse Proportion**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

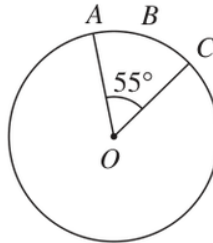
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jan 2022 P1H - Jan 2022 | P1H

16 The diagram shows a circle with centre O Diagram **NOT**
accurately drawn A , B and C are points on the circle so that the length of the arc ABC is 5 cm.Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm²

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16

Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

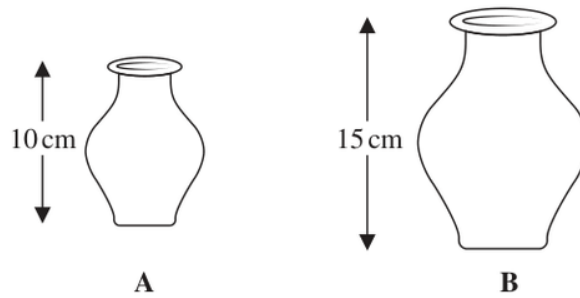
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jan 2022 P1H - Jan 2022 | P1H

17 A and B are two similar vases.Diagram **NOT**
accurately drawnVase **A** has height 10 cm.Vase **B** has height 15 cm.The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3 Calculate the volume of vase **A**..... cm^3

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2022 P1H - Jan 2022 | P1H

18 $A = w - \frac{x^2}{y}$

$w = 3.45$ correct to 2 decimal places.

$x = 1.9$ correct to 1 decimal place.

$y = 5$ correct to the nearest whole number.

Work out the lower bound of the value of A

Show your working clearly.

(Total for Question 18 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 18

Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Simultaneous Equations**

Jan 2022 P1H - Jan 2022 | P1H

19 Solve the simultaneous equations

$$\begin{aligned} 3x^2 + y^2 - xy &= 5 \\ y &= 2x - 3 \end{aligned}$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 19

Marks: 5

TOPIC: **Simultaneous Equations**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Completing the Square**

Jan 2022 P1H - Jan 2022 | P1H

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$
The point **A** is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of **A**

(..... ,)
(1)

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Completing the Square**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Completing the Square**

Jan 2022 P1H - Jan 2022 | P1H

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$
The point **A** is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of **A**

(..... ,)
(1)

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 20

Marks: 4

TOPIC: **Completing the Square**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

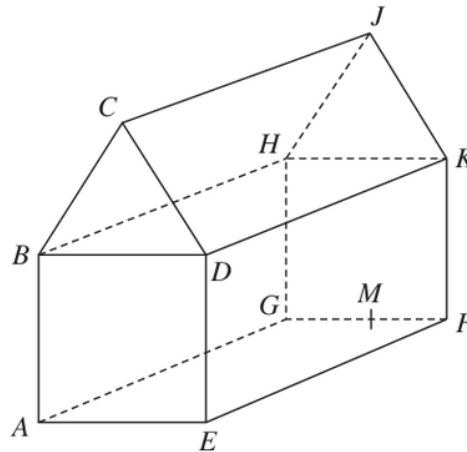
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$ Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$$T = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

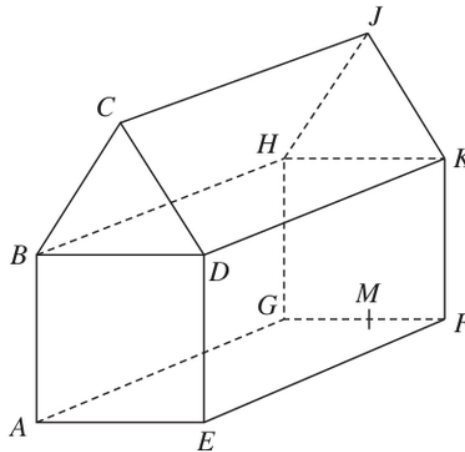
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$ Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$$T = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

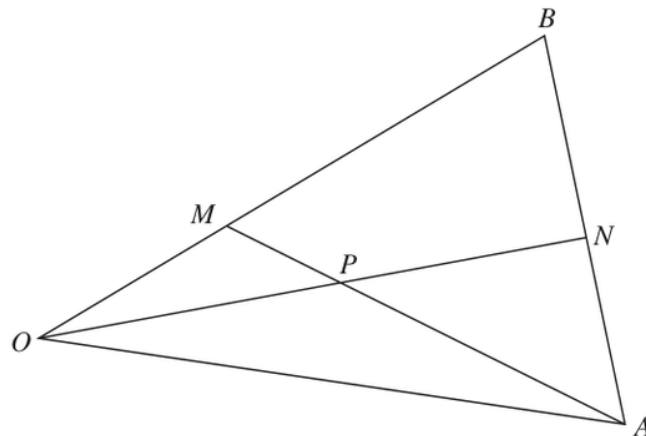
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Vectors**

Jan 2022 P1H - Jan 2022 | P1H

22 The diagram shows triangle OAB Diagram **NOT**
accurately drawn

$$\vec{OA} = 8\mathbf{a} \quad \vec{OB} = 6\mathbf{b}$$

 M is the point on OB such that $OM : MB = 1 : 2$ N is the midpoint of AB P is the point of intersection of ON and AM

Using a vector method, find $\vec{OP}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$
 Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Vectors**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

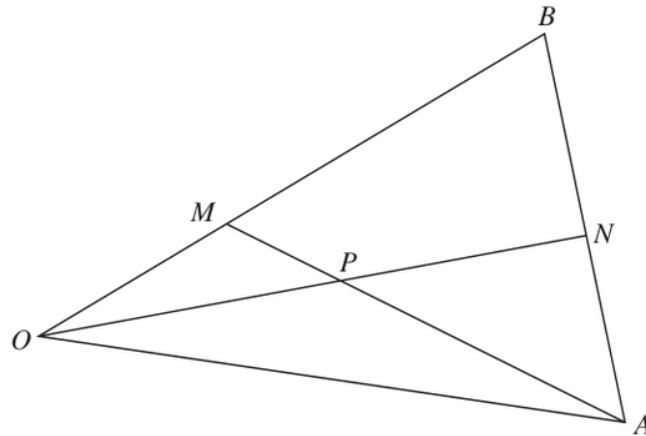
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Vectors**

Jan 2022 P1H - Jan 2022 | P1H

22 The diagram shows triangle OAB Diagram **NOT**
accurately drawn

$$\vec{OA} = 8\mathbf{a} \quad \vec{OB} = 6\mathbf{b}$$

 M is the point on OB such that $OM:MB = 1:2$ N is the midpoint of AB P is the point of intersection of ON and AM

Using a vector method, find $\vec{OP}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$
Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Vectors**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

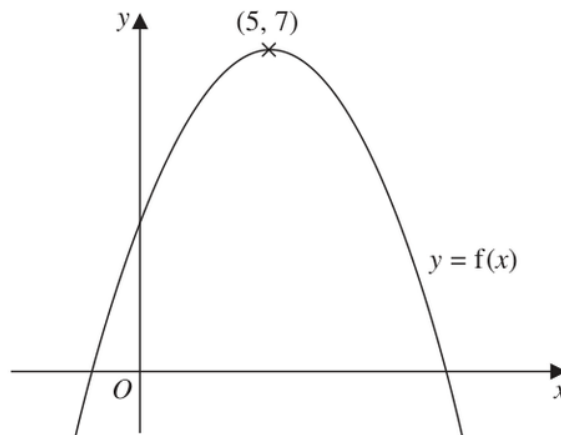
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 2****TOPIC: Transformations of Graphs**

Jan 2022 P1H - Jan 2022 | P1H

23 The diagram shows a sketch of the curve with equation $y = f(x)$ 

There is only one maximum point on the curve.
 The coordinates of this maximum point are (5, 7)

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 9)$

(.....,)

(ii) $y = f(x) + 3$

(.....,)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 2

TOPIC: **Transformations of Graphs**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

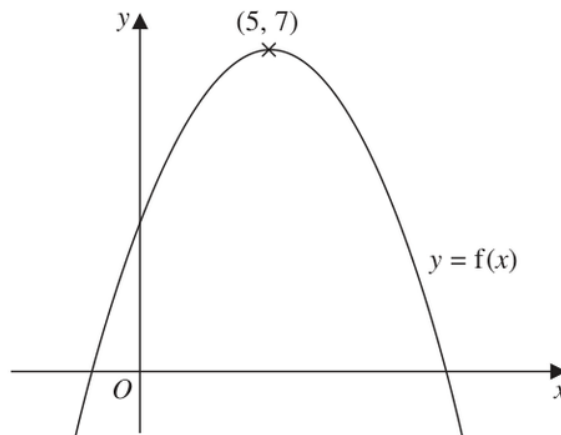
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 2****TOPIC: Transformations of Graphs**

Jan 2022 P1H - Jan 2022 | P1H

23 The diagram shows a sketch of the curve with equation $y = f(x)$ 

There is only one maximum point on the curve.
 The coordinates of this maximum point are (5, 7)

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 9)$

(.....,)

(ii) $y = f(x) + 3$

(.....,)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 2

TOPIC: **Transformations of Graphs**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Differentiation**

Jan 2022 P1H - Jan 2022 | P1H

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

(Total for Question 24 is 6 marks)

EA

PAST PAPER SOLUTION

Q#: 24

Marks: 6

TOPIC: **Differentiation**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Differentiation**

Jan 2022 P1H - Jan 2022 | P1H

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

(Total for Question 24 is 6 marks)

EA

PAST PAPER SOLUTION

Q#: 24

Marks: 6

TOPIC: **Differentiation**

Jan 2022 P1H - Jan 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA